

1: Configure AWS EC2 instance. (Windows Server) Ensured that each Windows host is:

- Running and reachable via public/private IP.
- Configured with WinRM (HTTP) and a user (ansibleadmin) with admin privileges.
- Has firewall rules allowing WinRM traffic (default ports: 5985 for HTTP, 5986 for HTTPS).

2: Enable WinRM on Windows Hosts.

Use PowerShell (ISE)

```
winrm quickconfig -q
```

```
winrm set winrm/config/service/auth '@{Basic="true"}'
```

```
winrm set winrm/config/service '@{AllowUnencrypted="true"}'
```

```
Set-Item -Path WSMAN:\localhost\Service\AllowUnencrypted -Value $true
```

Create a new local user

```
New-LocalUser -Name "ansibleadminu" -Password (Read-Host -AsSecureString "Enter Password") -FullName "Ansible User" -Description "User for Ansible management"
```

Add the user to Administrators group (optional, for elevated rights)

```
Add-LocalGroupMember -Group "Administrators" -Member "ansibleadminu"
```

3: Install Required Ansible Collections and python packages on local system:

a:) Install Required Python Package on Control Node On control node:

```
pip install "pywinrm>=0.3.0"
```

```
sudo dnf install python3 python3-pip -y
```

(Optional) Install Kerberos support If you plan to use Kerberos authentication (not required for basic auth):

```
pip3 install pywinrm[kerberos]
```

Verify Installation Run this to confirm:

```
python3 -c "import winrm; print(winrm.__version__)"
```

You should see the version number (e.g., 0.4.3).

 **Pro Tip for RHEL If you're using a virtual environment for Ansible:**

```
python3 -m venv ansible-env
```

```
source ansible-env/bin/activate
```

```
pip install "pywinrm>=0.3.0"
```

This keeps your system Python clean and avoids conflicts

b:) Install ansible collections:

```
ansible-galaxy collection install ansible.windows
```

```
ansible-galaxy collection install community.windows
```

4: Create Inventory for AWS Windows hosts

```
[awswindows]
```

```
98.81.53.254
```

```
98.81.69.225
```

```
[awswindows:vars]
```

```
ansible_user=ansibleadmin
```

```
ansible_password="{{ vault_win_password }}"
```

```
ansible_connection= winrm
```

```
ansible_winrm_transport=basic
```

```
ansible_winrm_server_cert_validation=ignore #for production use HTTPS
```

```
ansible_port=5985
```

5: Secure Credentials with Ansible Vault.

```
ansible-vault create group_vars/awswindows.yml
```

```
vault_win_password: "YourSecurePassword"
```

6: Try to ping awswindows hosts.

Make sure firewall allow or off at EC2-Instance

```
ansible awswindows -m win_ping --ask-vault-pass
```

7:create a play-book.

- name: Configure Windows Hosts

hosts: awswindows

gather_facts: no #facts gathering Disabled.

tasks:

- name: Ensure IIS is installed

win_feature:

name: Web-Server

state: present

- name: Create a directory

win_file:

path: C:\Tools

state: directory

#Run playBook

ansible-playbook aws.yml --ask-vault-pass

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